Examples:

a)	$\frac{(-3)^5}{(-3)^2} = (-3)^{5-2}$ by law and $(-3)^3 = -27$ (or) $\frac{(-3)^5}{(-3)^2} = (-3)^5 \times (-3)^{-2} = (-3)^{5-2} = (-3)^3$ (or) $\frac{(-3)^5}{(-3)^2} = \frac{(-3)\times(-3)\times(-3)\times(-3)\times(-3)}{(-3)\times(-3)} = (-3)\times(-3)\times(-3) = (-3)^3$
b)	$\frac{(-7)^{100}}{(-7)^{98}} = (-7)^{100-98}$ by law and $(-7)^2 = 49$ (or) $\frac{(-7)^{100}}{(-7)^{98}} = \frac{(-7)\times(-7)\times(-7)\times \ldots 100 \text{ times}}{(-7)\times(-7)\times(-7)\times \ldots 98 \text{ times}} = (-7)\times(-7) = 49$

Power law

According to this law, when raising a power to another power, we can just multiply the exponents.

$$(a^m)^n = a^{mn}$$

where a ($a \neq 0$), m, n are integers.

Examples:

$[(-2)^3]^2 = (-2)^{3\times2}$ by law and $(-2)^6 = 64$

(or)

$[(-2)^3]^2 = [(-2)\times(-2)\times(-2)]^2 = [-8]^2 = 64$



Try these

Verify the following rules (as we did above). Here, a,b are non-zero integers and m, n are any integers.

1. Product of same powers to power of product rule: $a^m \times b^m = (ab)^m$
2. Quotient of same powers to power of quotient rule: $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$
3. Zero exponent rule: $a^0 = 1$.

Example 1.36

Find the value of (i) 4^{-3} (ii) $\frac{1}{2^{-3}}$ (iii) $(-2)^5 \times (-2)^{-3}$ (iv) $\frac{3^2}{3^{-2}}$

Solution:

(i) $4^{-3} = \frac{1}{4^3} = \frac{1}{4\times4\times4} = \frac{1}{64}$ (ii) $\frac{1}{2^{-3}} = 2^3 = 2\times2\times2 = 8$

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